

ES 193: Homework 2

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Part 1: Tasks

```
# library function loads in packages
library(tidyverse)
library(janitor)
library(effectsize)
```

```
# we are bringing in the urchin-expirement-data.csv
# using the read_csv function and storing it as an object called 'urchins'
urchins <- read_csv("urchin-experiment-data.csv")
```

```
# bringing in screen_time.csv data and storing it as my_data
my_data <- read_csv("screen_time.csv")
```

Part 2: Problems

Problem 1. Raptors at the bird refuge

a.

I collected discrete data since I am collecting a whole counts of Red-tailed Hawks present. You cannot count fractions of alive Red-Tailed Hawks.

b.

Standard deviation is a better descriptor of the Red-tailed Hawk count as it is the average distance observations are from the mean

```
# creating object with hawk count values
hawk_count <- c(1,3,2,0,1,1,3,2,0,0,1,1)

# display data frame
hawk_count
```

```
[1] 1 3 2 0 1 1 3 2 0 0 1 1
```

```
# calculating standard deviation of the hawk count sample
sd_hawk <- sd(hawk_count)

# display hawk count standard deviation
sd_hawk
```

```
[1] 1.05529
```

The standard deviation of the Red-tailed Hawk sample data is 1.1 Red-tailed Hawks.

c.

Standard error is a better descriptor of the precision of the Red-tailed Hawk count as it quantifies how much the sample mean deviates from the true population mean of Red-tailed Hawks.

```
# calculating hawk standrad error
se_hawk <- sd_hawk/sqrt(length(hawk_count))

# display result of se_hawk
se_hawk
```

```
[1] 0.3046359
```

The standard error of the Red-tailed Hawk sample data is 0.3 Red-tailed Hawks.

Problem 2. Urchin gonad weights

a.

```

# we are going to make changes to the urchins data and
# store it as urchins_clean object
urchins_clean <- urchins |>
  # use clean_names function to make the column names lower case
  # and underscores
  clean_names() |>
  # mutate column names to change the c and n designations to be
  # Control and Nutrient-enriched
  mutate(experimental_treatment = case_when(
    experimental_treatment == "c" ~ "Control",
    experimental_treatment == "n" ~ "Nutrient-enriched"
  )) |>
  select(experimental_treatment, weight_g) |>
  # drop any na observations
  drop_na(weight_g)

# display 10 random rows from the urchins_clean data frame
# using slice_sample()
slice_sample(urchins_clean,
  n = 10,
  by = NULL,
  weight_by = NULL,
  replace = FALSE)

```

```

# A tibble: 10 x 2
  experimental_treatment weight_g
  <chr>                <dbl>
1 Control              37
2 Control             21.2
3 Nutrient-enriched   33.2
4 Control              6.1
5 Nutrient-enriched   31.8
6 Nutrient-enriched   33.3
7 Nutrient-enriched   26
8 Control             17.6
9 Control             25.4
10 Control            33.8

```

b.

```
# store urchins_clean summary data as the object urchins_summary
urchins_summary <- urchins_clean |>
  # group by the treatment type
  group_by(experimental_treatment) |>
  # begin summarizing both treatment types
  summarise(
    # calculate mean gonad weight
    mean_weight = mean(weight_g),
    # summarise number of observations per treatment type
    n = n(),
    # calculate standard deviation for gonad weight
    sd_weight = sd(weight_g)
  )

# display urchins_summary data frame
urchins_summary
```

```
# A tibble: 2 x 4
  experimental_treatment mean_weight      n sd_weight
  <chr>                  <dbl> <int>    <dbl>
1 Control                26.8    42     8.67
2 Nutrient-enriched      31.7    35     3.91
```

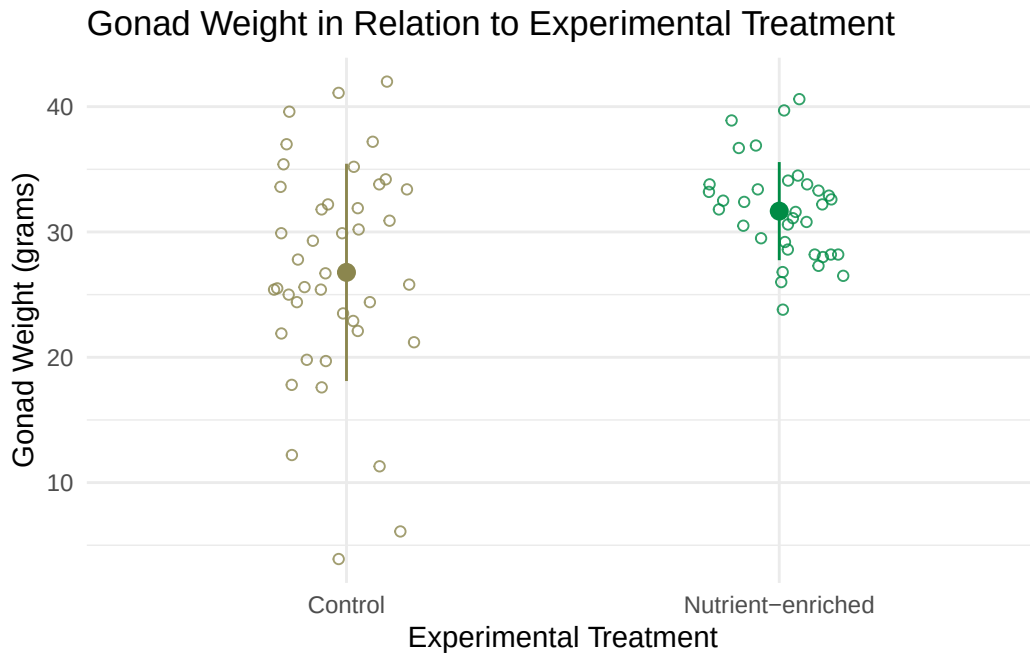
c.

```
# base layer is ggplot
# data is urchins_clean
ggplot(data = urchins_clean,
  # experimental treatment on the x axis
  # gonad weight (grams) on the y axis
  # coloring the points based on experimental treatment type
  mapping = aes(x = experimental_treatment,
    y = weight_g,
    color = experimental_treatment
  )) +
# first layer is a jitter plot
geom_jitter(
  # width is set to 0.15 to tighten the horizontal jitter
  width = 0.17,
  # height is set to 0 so that there is no vertical jitter
  height = 0,
  # open circle point shape for jitter points
```

```

        shape = 21,
        # increasing transparency so that it is easier to see the mean
        alpha = 0.8
    ) +
# adding a point that is the mean weight of each group
# using urchins_summary data
geom_pointrange(data = urchins_summary,
               # points displayed based on experimental treatment type
               mapping = aes(x = experimental_treatment,
                             # mean weight is the y value
                             y = mean_weight,
                             # upper bound is the mean plus
                             # a standard deviation
                             ymax = mean_weight + sd_weight,
                             # lower bound is the mean minus
                             # a standard deviation
                             ymin = mean_weight - sd_weight)) +
# manually setting colors for the treatment groups
scale_color_manual(values = c("Control" = "khaki4",
                              "Nutrient-enriched" = "springgreen4")) +
# changing axis labels and title
labs(x = "Experimental Treatment",
     y = "Gonad Weight (grams)",
     title = "Gonad Weight in Relation to Experimental Treatment"
    ) +
# changing theme from default theme to theme_minimal
theme_minimal() +
# removing the legend
theme(legend.position = "none")

```



d.

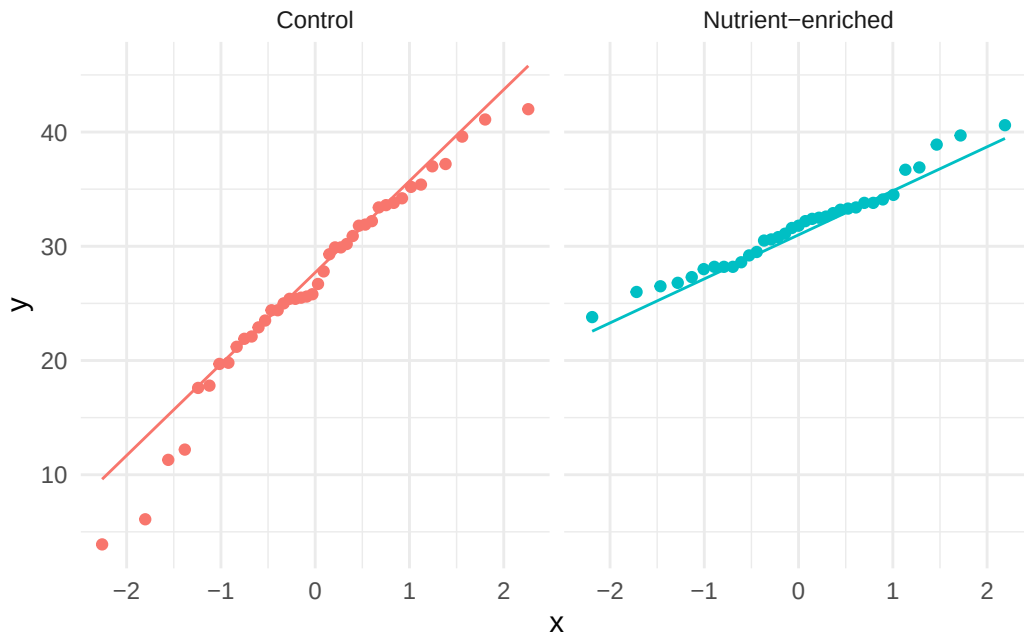
H[0]: There is no difference in mean urchin gonad weight between the Control group (mean = 26.8), and the Nutrient-enriched group.

H[A]: There is a difference in mean urchin gonad weight between the Control group (mean = 26.8), and the Nutrient-enriched group.

e.

```
# base layer is ggplot
ggplot(data = urchins_clean,
       # set sample as weight
       aes(sample = weight_g,
           # set color based on treatment
           color = experimental_treatment)) +
# first layer is qq plot
geom_qq() +
# second layer is qq line
geom_qq_line() +
# separate the two treatments into separate graphs
facet_wrap(~experimental_treatment) +
# change theme from default
theme_minimal() +
```

```
# remove legend
theme(legend.position = "none")
```



f.

The gonad weight (g) variable is normally distributed as the points remain close to the qq line throughout most of the line with just some tail offs at the ends.

g.

Looking at the QQ plot I would say that the variances are pretty similar but the control group likely has more variance due to its outliers at the lower tail end.

```
# create object called var_gonad_weight to store variances for each group
var_gonad_weight <- urchins_clean |>
  # group variance calculation by treatment
  group_by(experimental_treatment) |>
  # use summarise function and calculate variance
  summarise(var = var(weight_g)) |>
  # ungroup data frame
  ungroup()

# calculate F test manually with outputted variances
75.2/15.3
```

[1] 4.915033

```
# F test of equal variance function
var.test(
  # comparing variance of weight_g by experimental_treatment group
  weight_g ~ experimental_treatment,
  # using urchins_clean data
  data = urchins_clean
)
```

F test to compare two variances

```
data: weight_g by experimental_treatment
F = 4.9054, num df = 41, denom df = 34, p-value = 7.185e-06
alternative hypothesis: true ratio of variances is not equal to 1
95 percent confidence interval:
 2.527793 9.330193
sample estimates:
ratio of variances
      4.905405
```

Ratio of variances is 4.9 indicating that the variances are not equal between the groups

h.

```
# t.test is the t-test function in R
t.test(
  # weight_g is the response variable based on the
  # experimental treatment group
  weight_g ~ experimental_treatment,
  # using urchins_clean data
  data = urchins_clean,
  # to run a Welch's t-test of unequal variance, set variance equal to FALSE
  var.equal = FALSE
)
```

Welch Two Sample t-test

```
data: weight_g by experimental_treatment
```

```

t = -3.2739, df = 59.238, p-value = 0.001774
alternative hypothesis: true difference in means between group Control and group Nutrient-en
95 percent confidence interval:
  -7.873119 -1.900214
sample estimates:
      mean in group Control mean in group Nutrient-enriched
      26.77619              31.66286

```

i.

I used a t-test since my hypothesis is comparing the mean of two groups.

I evaluated normality with a QQ plot and checked the homogeneity of variance using an F test.

I decided to use a Welch's t-test of unequal variance since the F test result demonstrated a large difference in variances among the two treatment groups.

j.

```

# performing a Cohen's d analysis
# cohens_d is the function in R
cohens_d(
  # weight_g is the response variable,
  # experimental_treatment is the predictor variable
  weight_g ~ experimental_treatment,
  # using urchins_clean data
  data = urchins_clean,
  # standard deviations are not pooled
  # since the two groups have unequal variances
  pooled_sd = FALSE
)

```

Cohen's d	95% CI
-0.73	[-1.18, -0.27]

- Estimated using un-pooled SD.

k.

We found evidence to suggest that nutrient-enriched seaweed has a moderate effect (Cohen's $d = -0.73$) on mean gonad weight (g) between urchins fed the nutrient-enriched seaweed and urchins fed normal seaweed (Welch's t-test, $t(59.24) = -3.27$, $p = 0.0017$, $\alpha = 0.05$).

On average, urchins fed nutrient-enriched seaweeds ($n = 35$, mean = 31.66) had a mass 4.88 grams greater (95% confidence interval: [-7.87, -1.9]) than urchins fed non-enriched seaweed ($n = 42$, mean = 26.78).

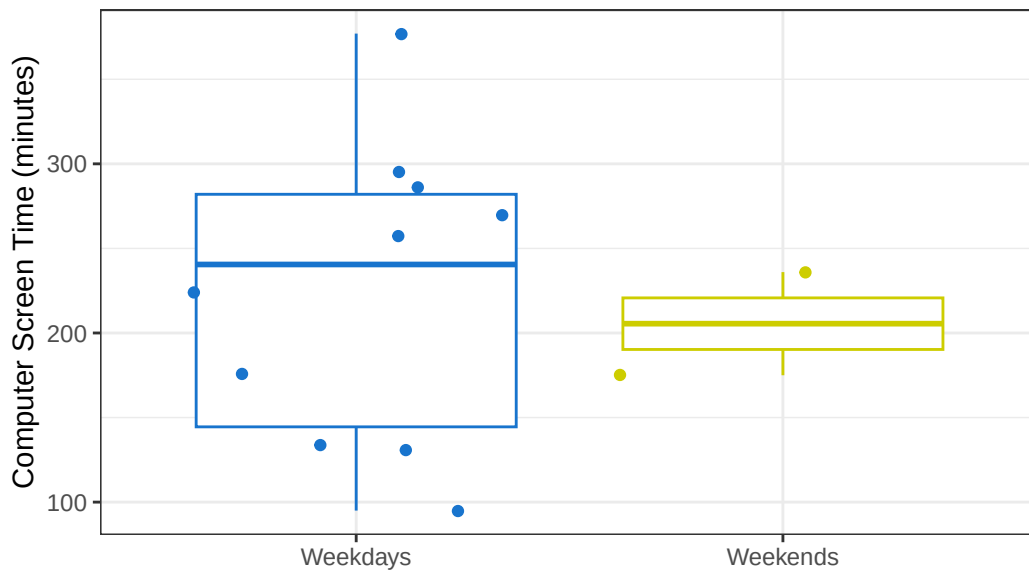
Problem 3

```
# cleaning my_data and storing it as object my_data_clean
my_data_clean <- my_data |>
  # dropping rows that contain NA values in the computer_time column
  drop_na(computer_time) |>
  # changing weekday variable inputs
  mutate(weekday = case_when(
    # y inputs in the weekday column are now displayed as Weekdays
    # n inputs in the weekday column are not displayed as Weekends
    weekday == "y" ~ "Weekdays",
    weekday == "n" ~ "Weekends"
  ))
```

a.

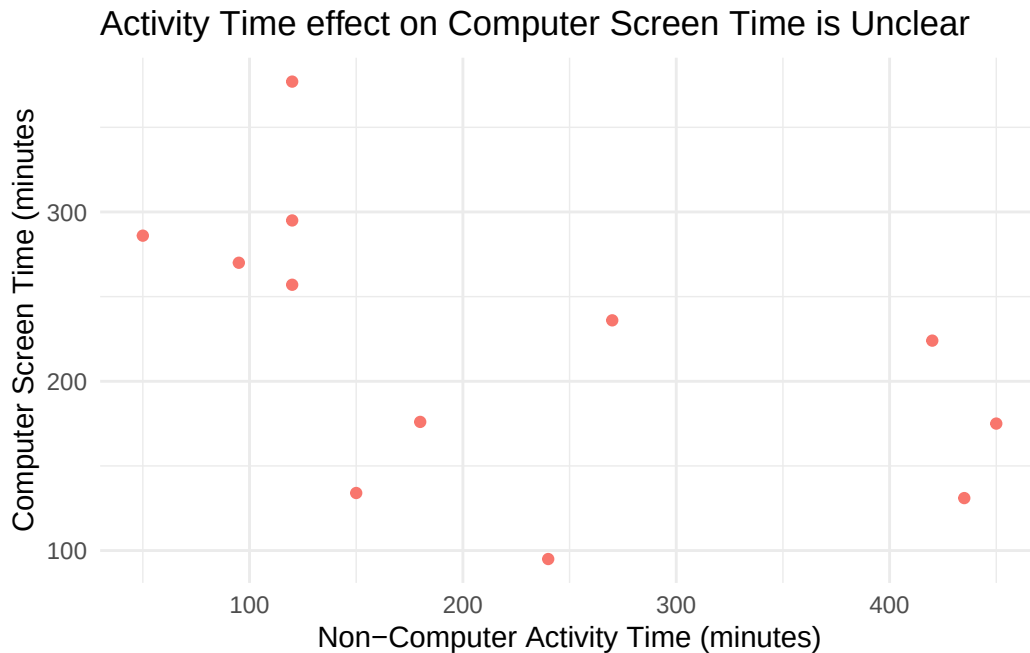
```
# base layer is ggplot
# data is my_data_clean
ggplot(data = my_data_clean,
  # x axis is whether it is a weekday or weekend
  # y axis is the computer screen time
  # color is based on whether it is the weekday or weekend group
  mapping = aes(x = weekday,
    y = computer_time,
    color = weekday)) +
  # boxplot is second layer
  geom_boxplot() +
  # jitter plot is the third layer
  geom_jitter() +
  # renaming y axis, x axis, and title
  labs(y = "Computer Screen Time (minutes)",
    x = "",
    title = "Weekdays Have Greater Spread in Computer Screen Time") +
  # manually setting color for Weekdays and Weekends groups
  scale_color_manual(values = c("Weekdays" = "dodgerblue3",
    "Weekends" = "yellow3")) +
  # changing the theme from the default
  theme_bw() +
  # removing legend
  theme(legend.position = "none")
```

Weekdays Have Greater Spread in Computer Screen Time



b.

```
# ggplot is base layer
ggplot(data = my_data_clean,
  # x axis is time spent on non computer activities
  # y axis is computer screen time
  # the color of the plot will be orchid4
  mapping = aes(x = time_activities,
    y = computer_time,
    color = "orchid4")) +
# points are second layer
geom_point() +
# renaming x axis, y axis, and title
labs(x = "Non-Computer Activity Time (minutes)",
  y = "Computer Screen Time (minutes)",
  title = "Activity Time effect on Computer Screen Time is Unclear") +
# changing theme from the default
theme_minimal() +
# removing legend
theme(legend.position = "none")
```



c.

Based on the boxplot figure of my data thus far, it appears that my mean screen time may be higher on the weekdays vs the weekends. In addition, the Weekdays box plot has more spread than the Weekends boxplot which you can see in the interquartile range box and the longer whiskers. More data may increase the spread of the Weekends Screen Time observations as there is currently only two observations. In addition, the difference in Computer Screen Time means may become more clear and distinct between the two groups, or become more similar with more data. The graph displaying Non-Computer Activity Time vs Computer Screen Time does not demonstrate any clear trend thus far.

d.

It was pretty easy to get my data from Google Sheets into R. My system of collecting data worked as I named all my columns using lowercase letters and underscores to space words. The data was already pretty clean and I just removed the rows that had NA values for computer_time. I did not remove rows with NA values in the notes and notes_activities column as it is not necessary to have notes for each day to make it a valid observation.

Problem 4

a.

I am interested in this paper because I am passionate about habitat restoration and was curious as to how professionals in the field factor in climate change. I also attended a zoom presentation of it so I have more of a human connection with the researchers.

b.

Wasson and contributors are addressing their hypothesis of how successfully plants will establish after the soil topography has been formed and the project has begun. There are three possible trajectories that they discuss, early success, where plants establish effectively in the first years, steady, where success is relatively similar through time, and late success. The researchers hypothesized that the Elkorn Slough would be characterized by the early success model.

The researchers also address five predictions which have to do with how success of plant establishment will change over time. For example, one of the predictions is that the establishment rate of new seedlings will decrease over time.

c.

I have selected one of the Wilcoxon's test.

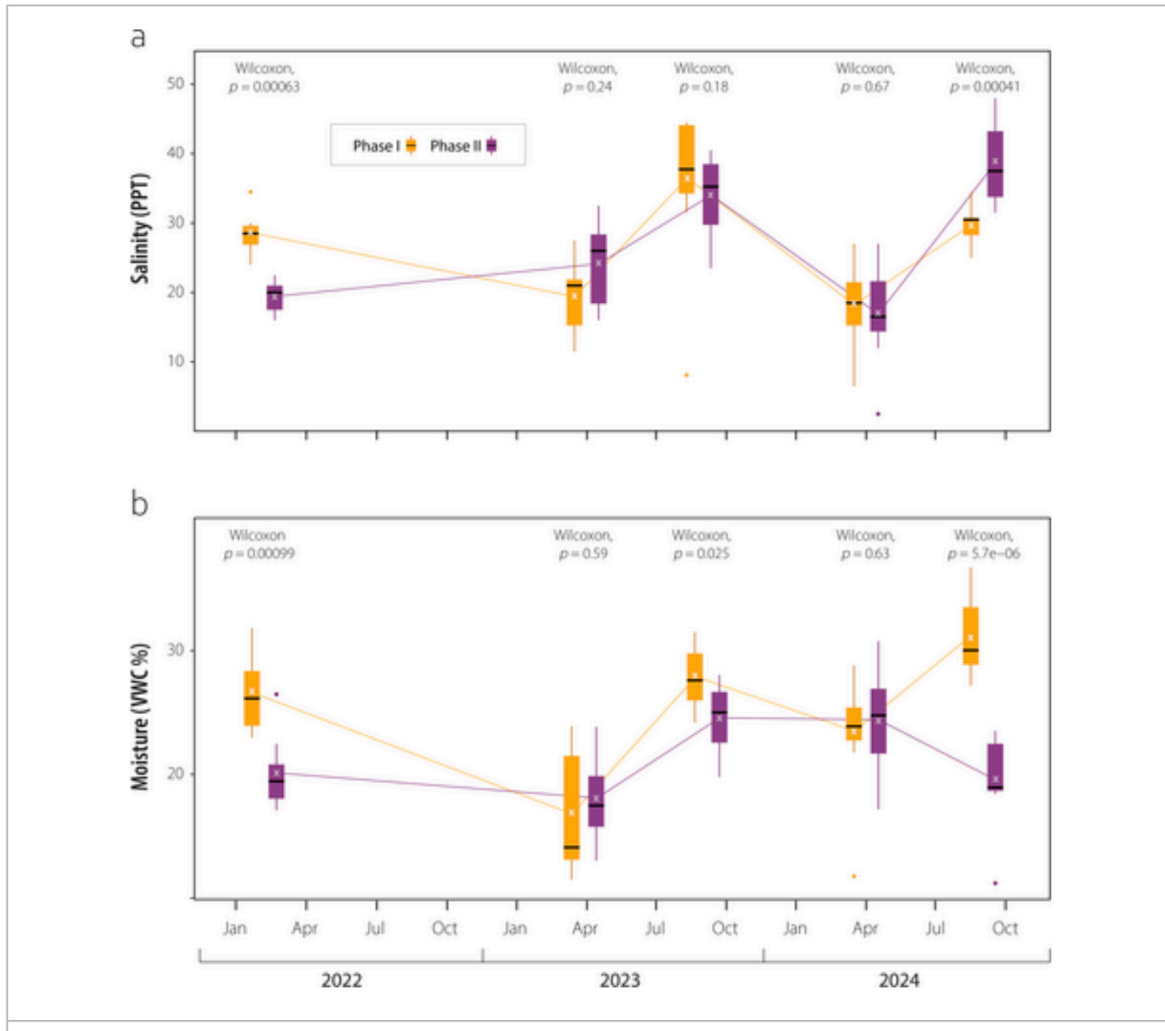
The response variables are soil salinity and water content. The predictor variable is one of the two Phase sites.

d.

The Wilcoxon test addresses the authors main question about the difference of soil abiotic factors between the two Phase sites.

A significant result would indicate that there is a difference in soil conditions between the two phase sites and thus they can use the two sites to compare how soil salinity and soil moisture affect plant establishment success.

e.



f.

The top graph displays time on the x axis, and soil salinity on the y axis. The figure shows how soil salinity varies throughout the seasons and how salinity differs between sites. The main message is that soil salinity was initially higher at the Phase I site, and then at the last measurements showed Phase II having higher soil salinity likely due to its higher elevation.

The bottom graph displays time on the x axis and soil moisture on the y axis. The figure shows how soil moisture was higher at the Phase I site during February 2022, September 2023, and September 2024.